

Mercury exposure and risks from dental amalgam in the US population, post-2000.



Dental amalgam is 50% metallic mercury (Hg) by weight and Hg vapour continuously evolves from in-place dental amalgam, causing increased Hg content with increasing amalgam load in urine, faeces, exhaled breath, saliva, blood, and various organs and tissues including the kidney, pituitary gland, liver, and brain. The Hg content also increases with maternal amalgam load in amniotic fluid, placenta, cord blood, meconium, various foetal tissues including liver, kidney and brain, in colostrum and breast milk. Based on 2001 to 2004 population statistics, 181.1 million Americans carry a grand total of 1.46 billion restored teeth. Children as young as 26 months were recorded as having restored teeth. Past dental practice and recently available data indicate that the majority of these restorations are composed of dental amalgam. Employing recent US population-based statistics on body weight and the frequency of dentally restored tooth surfaces, and recent research on the incremental increase in urinary Hg concentration per amalgam-filled tooth surface, estimates of Hg exposure from amalgam fillings were determined for 5 age groups of the US population. Three specific exposure scenarios were considered, each scenario incrementally reducing the number of tooth surfaces assumed to be restored with amalgam. Based on the least conservative of the scenarios evaluated, it was estimated that some 67.2 million Americans would exceed the Hg dose associated with the reference exposure level (REL) of 0.3 $\mu\text{g}/\text{m}^3$ established by the US Environmental Protection Agency; and 122.3 million Americans would exceed the dose associated with the REL of 0.03 $\mu\text{g}/\text{m}^3$ established by the California Environmental Protection Agency. Exposure estimates are consistent with previous estimates presented by Health Canada in 1995, and amount to 0.2 to 0.4 $\mu\text{g}/\text{day}$ per amalgam-filled tooth surface, or 0.5 to 1 $\mu\text{g}/\text{day}/\text{amalgam-filled tooth}$, depending on age and other factors. Copyright © 2011 Elsevier B.V. All rights reserved.